① CSE 321 1705071

CT 3

Answer

In OSI layers,
the Physical layer is the sole layer to actually send the bits through the transmission line. So, this layer is needed

The data link layer ensures that the next node in the topology gets the data by recieving acks or any other way Since, this representation enforces the condition, this layer is needed. However, this layer has to check that the sender's id is his own to stop transmitting further

The network layer deals with choosing a route. But here is only one route in each transmission. So, routing not needed. Network layer also does flow control or congestion control which is also unnecessary here. So, this layer is not needed.

The transport layer ensures the data reaches in end device. (not needed here) Does error control, packet reordering on so on. So, this layer is not needed

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These functionalities mentioned here can be done with just application layer (needed).

Session and presentation layer not needed.

So, the stack is

Application Layer

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Data Link layer (with extra checking)

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Physical Layer.

Now, fiber optics and copper wire these are hardwares that deal with the data rate of the channel through bandwidth. They are connected to transmission media and have no control over protocol stack. So, both stacks will be same.